Ankit Agrawal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Gurukul Science Academy	Class XII (Rajasthan)	2019	95.60%
Kamla Bharati Shikshan Sansthan School	Class X (Rajasthan)	2017	92.17%

EXPERIENCE

National Technical Research Organisation (NTRO)

Research Intern - AI/ML & Data Analysis Tooling

- Developed an offline data-analysis workflow integrating Python, FastAPI, and **CyberChef** to execute user-defined encoding and decoding operations through a local backend.
- Developed and evaluated **DNN** and **LightGBM** classifiers for distinguishing compressed and encrypted data, achieving **98.27% accuracy** on the binary text classifier and **100% classification Accuracy** on an initial JPEG-compressed vs. AES-encrypted image dataset.
- Implemented **NIST statistical randomness tests**, engineered entropy-, frequency-, and statistical features, and built a PySide6 GUI integrating ML models and analysis tests for offline evaluation.

PROJECTS

Deep Learning-Based Cryptanalysis of Feistel Networks

- Developed neural differential distinguishers for 128-bit Feistel ciphers including SM4, Camellia, and CLEFIA using **ResNet**, **DBitNet**, and **Inception** architectures.
- Built balanced datasets and evaluation pipelines for **neural distinguishers**, with automated input-difference selection, model training, and evaluation using accuracy, TPR, and TNR.
- Achieved **100% distinguishing accuracy** up to 7 rounds for the TNT-SM4 Scheme 02 construction.

Anti-Emulation Detection via Synthetic Sensor Data

- Processed real accelerometer, gyroscope, and magnetometer data by aligning timestamps, handling missing values, standardizing sampling to 100 Hz, and generating a 19-feature multi-sensor dataset.
- Evaluated **TimeGAN**, **RCGAN**, SMOTE, **VAR**, data augmentation, and PSD-based synthesis for generating realistic sensor sequences while preserving temporal and cross-sensor characteristics.
- Compared model performance across the evaluated datasets, with VAR outperforming the other evaluated approaches overall in the project analysis

Secure Implementation of a Web Application

- Developed a secure web platform utilizing HTML, CSS, Bootstrap, and PHP, actively defending against **SQL Injection**, **XSS**, and **IDOR vulnerabilities**.
- Managed collaborative development across multi-branch Git repositories, successfully resolving complex merge conflicts to integrate secure code updates from project teammates.

LLM From Scratch -- Hands-on Implementation

- Implemented core **LLM preprocessing and input-representation components** in Python, including tokenization, **Byte Pair Encoding (BPE)**, token IDs, context tokens, sliding-window sampling, token embeddings, and positional encoding.
- Developed hands-on understanding of **GPT-style LLM architecture**, progressing from data preparation toward attention mechanisms and model training.

HACKATHON

Detection of Backdoor Attacks using HPCs

- Developed a **one-class anomaly-detection pipeline** to identify backdoor neural-network inferences using **hardware performance counters (HPCs)**.
- Analyzed HPC correlations and evaluated **Mahalanobis-distance and Isolation Forest** based anomaly detection with clean-data validation and threshold analysis.

Cyber Security PSBs Hackathon 2026 -- Hackathon Submission

BioShield-AI: Adaptive Behavioral Authentication for Digital Banking

- Proposed a **continuous authentication framework** using keystroke, touch/kinetic, and device telemetry to identify anomalous user behavior.
- Designed a **hybrid AI approach** combining user-specific Autoencoders, 1D-CNN, and dynamic risk scoring for adaptive security responses.

SKILLS

- **Languages & Frameworks**: Python, C++, C, LATEX, Numpy, Pandas, SQL, Scikit-learn, Verilog
- **AI/ML & Data**: PyTorch, Tensorflow, CNNs, ResNet, Transformers, Time-Series Modeling
- **Engineering & Tools**: Git, Docker, Jupyter, Linux/Ubuntu, CyberChef, Automated Testing
- **Security & Analysis**: Malware Analysis, NIST Statistical Testing
- **Cryptanalysis**: Differential & Linear Attacks, DLCT, Security of Block Cipher, Neural Distinguisher
- **CyberSecurity:** Burp Suite, Sonic Visualiser, Audacity, Wireshark, Ghidra

POSITIONS OF RESPONSIBILITY

- PG Mentor at sunshine cell of IITH
- Core Member at CyberSecurity club(Kludge) of IITH